Getting Started with R

Data Analytics using R (textbook) - Chapter 2

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Working with Directory

`getwd()` - Get Current Working Directory

```r
getwd()
```

```
[1] "/home/anil/Documents/dsur26/programs"
```

`setwd()` - Set Working Directory

```r
setwd("/home/anil/Documents/dsur26/programs")
```

On Windows, use '/' or '\' as file separators.

`dir()` and `list.files()` Functions

```
dir(path = ".", pattern = NULL, all.files = FALSE, full.names = FALSE,
    recursive = FALSE, ignore.case = FALSE, include.dirs = FALSE)
```

```
 [1] "cha4-2-nb.Rmd"                    "chap2-1 introduction to R.Rmd"
 [3] "chap2-1-introduction-to-R.html" "chap2-1-introduction-to-R.log"
 [5] "chap2-1-introduction-to-R.tex"  "chap2-2.html"
 [7] "chap2-2.Rmd"                    "chap2-introduction -new.Rmd"
 [9] "chap2-introduction--new.html"   "chap2-introduction--new.Rmd"
[11] "chap3-nb.Rmd"                   "chap4-notebook.Rmd"
[13] "data_structures_in_r.webp"      "data_types89.webp"
[15] "database.R"                     "first-shiny.R"
[17] "lab-prog-dply-subset.html"      "lab-prog-dply-subset.Rmd"
[19] "mid-prog"                       "mid1R Notebook.Rmd"
[21] "myworkspace.RData"              "operators.webp"
[23] "RData"                          "shiny programs.nb.html"
[25] "shiny programs.Rmd"             "shiny-programs.html"
[27] "unit1-chap3.Rmd"
```

Example 1: List files in current directory

```
dir()
```

```
 [1] "cha4-2-nb.Rmd"                    "chap2-1 introduction to R.Rmd"
 [3] "chap2-1-introduction-to-R.html" "chap2-1-introduction-to-R.log"
 [5] "chap2-1-introduction-to-R.tex"  "chap2-2.html"
 [7] "chap2-2.Rmd"                    "chap2-introduction -new.Rmd"
 [9] "chap2-introduction--new.html"   "chap2-introduction--new.Rmd"
[11] "chap3-nb.Rmd"                   "chap4-notebook.Rmd"
[13] "data_structures_in_r.webp"      "data_types89.webp"
[15] "database.R"                     "first-shiny.R"
[17] "lab-prog-dply-subset.html"      "lab-prog-dply-subset.Rmd"
[19] "mid-prog"                       "mid1R Notebook.Rmd"
[21] "myworkspace.RData"              "operators.webp"
[23] "RData"                          "shiny programs.nb.html"
[25] "shiny programs.Rmd"             "shiny-programs.html"
[27] "unit1-chap3.Rmd"
```

Example 2: List files in a specific path

```
dir(path = "C:/Users/Seema_acharya")
```

```
character(0)
```

Advanced `dir()` Usage

Example 3: Full (absolute) paths

```r
dir(path = "/home/anil/Documents/dsur26/programs", full.names = TRUE)
```

```
 [1] "/home/anil/Documents/dsur26/programs/cha4-2-nb.Rmd"
 [2] "/home/anil/Documents/dsur26/programs/chap2-1 introduction to R.Rmd"
 [3] "/home/anil/Documents/dsur26/programs/chap2-1-introduction-to-R.html"
 [4] "/home/anil/Documents/dsur26/programs/chap2-1-introduction-to-R.log"
 [5] "/home/anil/Documents/dsur26/programs/chap2-1-introduction-to-R.tex"
 [6] "/home/anil/Documents/dsur26/programs/chap2-2.html"
 [7] "/home/anil/Documents/dsur26/programs/chap2-2.Rmd"
 [8] "/home/anil/Documents/dsur26/programs/chap2-introduction -new.Rmd"
 [9] "/home/anil/Documents/dsur26/programs/chap2-introduction--new.html"
[10] "/home/anil/Documents/dsur26/programs/chap2-introduction--new.Rmd"
[11] "/home/anil/Documents/dsur26/programs/chap3-nb.Rmd"
[12] "/home/anil/Documents/dsur26/programs/chap4-notebook.Rmd"
[13] "/home/anil/Documents/dsur26/programs/data_structures_in_r.webp"
[14] "/home/anil/Documents/dsur26/programs/data_types89.webp"
[15] "/home/anil/Documents/dsur26/programs/database.R"
[16] "/home/anil/Documents/dsur26/programs/first-shiny.R"
[17] "/home/anil/Documents/dsur26/programs/lab-prog-dply-subset.html"
[18] "/home/anil/Documents/dsur26/programs/lab-prog-dply-subset.Rmd"
[19] "/home/anil/Documents/dsur26/programs/mid-prog"
[20] "/home/anil/Documents/dsur26/programs/mid1R Notebook.Rmd"
[21] "/home/anil/Documents/dsur26/programs/myworkspace.RData"
[22] "/home/anil/Documents/dsur26/programs/operators.webp"
[23] "/home/anil/Documents/dsur26/programs/RData"
[24] "/home/anil/Documents/dsur26/programs/shiny programs.nb.html"
[25] "/home/anil/Documents/dsur26/programs/shiny programs.Rmd"
[26] "/home/anil/Documents/dsur26/programs/shiny-programs.html"
[27] "/home/anil/Documents/dsur26/programs/unit1-chap3.Rmd"
```

Example 4: Pattern matching – files starting with “D”

```r
dir(path = "/home/anil/Documents/dsur26/programs", pattern = "^m")
```

```
[1] "mid-prog"           "mid1R Notebook.Rmd" "myworkspace.RData"
```

Example 5: Recursive listing

```r
dir(path = "/home/anil/Documents/dsur26/programs", recursive = TRUE, include.dirs = TRUE)
```

```
 [1] "cha4-2-nb.Rmd"
 [2] "chap2-1 introduction to R.Rmd"
 [3] "chap2-1-introduction-to-R.html"
 [4] "chap2-1-introduction-to-R.log"
 [5] "chap2-1-introduction-to-R.tex"
 [6] "chap2-2.html"
 [7] "chap2-2.Rmd"
 [8] "chap2-introduction -new.Rmd"
 [9] "chap2-introduction--new.html"
[10] "chap2-introduction--new.Rmd"
[11] "chap3-nb.Rmd"
[12] "chap4-notebook.Rmd"
```

```
[13] "data_structures_in_r.webp"
[14] "data_types89.webp"
[15] "database.R"
[16] "first-shiny.R"
[17] "lab-prog-dply-subset.html"
[18] "lab-prog-dply-subset.Rmd"
[19] "mid-prog"
[20] "mid-prog/02Data.ppt"
[21] "mid-prog/03Preprocessing.ppt"
[22] "mid-prog/3factors and data frames.pdf"
[23] "mid-prog/3lists.pdf"
[24] "mid-prog/3matrices.pdf"
[25] "mid-prog/4conditional and control flow.pdf"
[26] "mid-prog/4functions in R.pdf"
[27] "mid-prog/4iterative programming.pdf"
[28] "mid-prog/base-r-cheat-sheet.pdf"
[29] "mid-prog/chap3 loading and handling data.Rmd"
[30] "mid-prog/chap3-loading-and-handling-data.html"
[31] "mid-prog/chap3-loading-and-handling-data.log"
[32] "mid-prog/chap3-loading-and-handling-data.tex"
[33] "mid-prog/DataScience_Notes-1-6.pdf"
[34] "mid-prog/example.db"
[35] "mid-prog/excel.html"
[36] "mid-prog/excel.Rmd"
[37] "mid-prog/IDA & lab DS minor degree .pdf"
[38] "mid-prog/images"
[39] "mid-prog/images/clipboard-2412255183.png"
[40] "mid-prog/images/data_structures_in_r.webp"
[41] "mid-prog/images/data_types89.webp"
[42] "mid-prog/images/operators.webp"
[43] "mid-prog/matrices.pdf"
[44] "mid-prog/mid-prog.Rproj"
[45] "mid-prog/mid1R Notebook.html"
[46] "mid-prog/mid1R Notebook.Rmd"
[47] "mid-prog/mid1R-Notebook.html"
[48] "mid-prog/mid1R-Notebook.log"
[49] "mid-prog/mid1R-Notebook.tex"
[50] "mid-prog/minor conditional-iterative.Rmd"
[51] "mid-prog/minor-conditional-iterative.html"
[52] "mid-prog/minor-functions.nb.html"
[53] "mid-prog/minor-functions.Rmd"
[54] "mid-prog/R18B.Tech.CSE(DataScience)IIIIVYearTentativeSyllabus.pdf"
[55] "mid-prog/slides for 1-2-3.pptx"
[56] "mid-prog/syllabus.pdf"
[57] "mid-prog/title-1.pdf"
[58] "mid-prog/unit 5.pdf"
[59] "mid-prog/unit1notes.docx"
[60] "mid1R Notebook.Rmd"
[61] "myworkspace.RData"
[62] "operators.webp"
[63] "RData"
[64] "shiny programs.nb.html"
[65] "shiny programs.Rmd"
[66] "shiny-programs.html"
[67] "unit1-chap3.Rmd"
```

`list.files()` is equivalent to `dir()` and accepts the same arguments.

Data Types in R

R Objects:

- Vector
- List
- Matrix
- Array
- Factor
- **Data Frames**

Basic Data Types

Type	Description	Example
Logical	TRUE / FALSE	TRUE, F
Numeric	Numbers	2, `76.25`
Integer	Whole numbers	`2L`
Character	Text strings	`"Data Science"`
Double	Double precision	`76.25`
Complex	Complex numbers	`5 + 5i`
Raw	Raw bytes	–

Logical Data Type

```
TRUE
```

```
[1] TRUE
```

```
class(TRUE)
```

```
[1] "logical"
```

```
T
```

```
[1] TRUE
```

```
class(F)
```

```
[1] "logical"
```

```
FALSE
```

```
[1] FALSE
```

```
class(FALSE)
```

```
[1] "logical"
```

T is shorthand for TRUE, F for FALSE.

Numeric Data Type

```r
2
```

```
[1] 2
```

```r
class(2)
```

```
[1] "numeric"
```

```r
76.25
```

```
[1] 76.25
```

```r
class(76.25)
```

```
[1] "numeric"
```

Integer Data Type

Note: Use the L suffix to create an integer. Integer is a subclass of numeric.

```
2L
```

```
[1] 2
```

```
class(2L)
```

```
[1] "integer"
```

```
is.numeric(2)
```

```
[1] TRUE
```

```
is.numeric(2L)
```

```
[1] TRUE
```

```
is.integer(2)
```

```
[1] FALSE
```

```
is.integer(2L)
```

```
[1] TRUE
```

Integers are numeric, but not all numbers are integers.

Character and Double Data Types

Character

```r
"Data Science"
```

```
[1] "Data Science"
```

```r
class("Data Science")
```

```
[1] "character"
```

```r
is.character("Data Science")
```

```
[1] TRUE
```

Double (double precision floating point)

```r
typeof(76.25)
```

```
[1] "double"
```

By default, numbers are “double” unless explicitly specified as integer with L.

Complex Data Type

```r
5 + 5i
```

```
[1] 5+5i
```

```r
class(5 + 5i)
```

```
[1] "complex"
```

Variables in R

```r
RectangleHeight <- 2
RectangleWidth  <- 4
RectangleArea   <- RectangleHeight * RectangleWidth
RectangleArea
```

```
[1] 8
```

List objects in environment

```r
ls()
```

```
[1] "RectangleArea"   "RectangleHeight" "RectangleWidth"
```

Clean environment

```r
rm(list = ls())
ls()
```

```
character(0)
```

Exploring the `mtcars` Dataset

Background

- Extracted from the 1974 *Motor Trend* US magazine.
- Fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

Variables in `mtcars`

Variable	Description
mpg	Miles/(US) gallon
cyl	Number of cylinders
disp	Displacement (cu.in.)
hp	Gross horsepower
drat	Rear axle ratio
wt	Weight (1000 lbs)
qsec	1/4 mile time
vs	V/S
am	Transmission (0 = auto, 1 = manual)
gear	Number of forward gears
carb	Number of carburetors

Loading the `mtcars` Dataset

```
installed.packages()
```

```
# Check installed packages


# Load the datasets package
library(datasets)

# Display the dataset
mtcars
```

Output (first few rows):

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

summary() Command

Provides a six-point summary (min, 1st quartile, median, mean, 3rd quartile, max) for each variable.

```
summary(mtcars)
```

```
      mpg             cyl             disp             hp       
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5  
 Median :19.20   Median :6.000   Median :196.3   Median :123.0  
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7  
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0  
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0  
      drat             wt             qsec             vs        
 Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
 1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000  
 Median :3.695   Median :3.325   Median :17.71   Median :0.0000  
 Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375  
 3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000  
 Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000  
       am              gear            carb      
 Min.   :0.0000   Min.   :3.000   Min.   :1.000  
 1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000  
 Median :0.0000   Median :4.000   Median :2.000  
 Mean   :0.4062   Mean   :3.688   Mean   :2.812  
 3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000  
 Max.   :1.0000   Max.   :5.000   Max.   :8.000  
```

`str()` Command – Structure

Displays the internal structure of an object.

Example 1: `str()` on itself

```
str(str)
```

```
function (object, ...)
```

Example 2: `str()` on `ls()`

```
str(ls)
```

```
function (name, pos = -1L, envir = as.environment(pos), all.names = FALSE,
    pattern, sorted = TRUE)
```

`str()` on `mtcars`

```
str(mtcars)
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

More `str()` Examples

Numeric vector

```r
x <- rnorm(100, 2, 4)
str(x)
```

```
 num [1:100] -0.987 5.738 2.759 6.592 5.448 ...
```

Matrix

```r
m <- matrix(rnorm(100), 10, 10)
str(m)
```

```
 num [1:10, 1:10] -1.389 -0.835 -0.275 -0.805 1.262 ...
```

```r
m[, 1]
```

```
 [1] -1.3888606 -0.8347800 -0.2748122 -0.8049208  1.2623070 -0.3963981
 [7]  0.7192626  0.1578967 -1.0349604 -0.2832729
```

`View()`, `head()`, `tail()`

`View()` – spreadsheet-like viewer

```
View(mtcars)
```

`head()` – first n observations (default 6)

```
head(mtcars, n = 6)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

head(x, n)

- x is the dataset or object.
- n specifies the number of rows to display.
- When n is a negative number, head() returns all rows except the last |n| rows.

`tail()` – last n observations (default 6)

```
tail(mtcars, n = 5)
```

```
                mpg cyl  disp  hp drat    wt qsec vs am gear carb
Lotus Europa   30.4   4  95.1 113 3.77 1.513 16.9  1  1    5    2
Ford Pantera L 15.8   8 351.0 264 4.22 3.170 14.5  0  1    5    4
Ferrari Dino   19.7   6 145.0 175 3.62 2.770 15.5  0  1    5    6
Maserati Bora  15.0   8 301.0 335 3.54 3.570 14.6  0  1    5    8
Volvo 142E     21.4   4 121.0 109 4.11 2.780 18.6  1  1    4    2
```

`ncol()`, `nrow()`, `edit()`, `fix()`

`ncol()` – number of columns

```
ncol(mtcars)
```

```
[1] 11
```

`nrow()` – number of rows

```
nrow(mtcars)
```

```
[1] 32
```

`edit()` – edit dataset (temporary, changes not saved)

```
edit(mtcars)
```

`fix()` – edit dataset (changes saved permanently)

```
fix(mtcars)
View(mtcars)  # see modifications
```

Command	Result
`head(mtcars)`	Displays the first 6 rows (default).
`head(mtcars, 10)`	Displays the first 10 rows.
`head(mtcars, -10)`	Displays all rows except the last 10 rows (22 rows).
`tail(mtcars)`	Displays the last 6 rows.
`tail(mtcars, -10)`	Displays all rows except the first 10 rows (22 rows).

data() Function

Lists all available datasets in loaded packages.

```
data()
```

Load the trees dataset

```
data(trees)
trees
```

```
   Girth Height Volume
1    8.3     70   10.3
2    8.6     65   10.3
3    8.8     63   10.2
4   10.5     72   16.4
5   10.7     81   18.8
6   10.8     83   19.7
7   11.0     66   15.6
8   11.0     75   18.2
9   11.1     80   22.6
10  11.2     75   19.9
11  11.3     79   24.2
12  11.4     76   21.0
13  11.4     76   21.4
14  11.7     69   21.3
15  12.0     75   19.1
16  12.9     74   22.2
17  12.9     85   33.8
18  13.3     86   27.4
19  13.7     71   25.7
20  13.8     64   24.9
21  14.0     78   34.5
22  14.2     80   31.7
23  14.5     74   36.3
24  16.0     72   38.3
25  16.3     77   42.6
26  17.3     81   55.4
27  17.5     82   55.7
28  17.9     80   58.3
29  18.0     80   51.5
30  18.0     80   51.0
31  20.6     87   77.0
```

Summary of trees

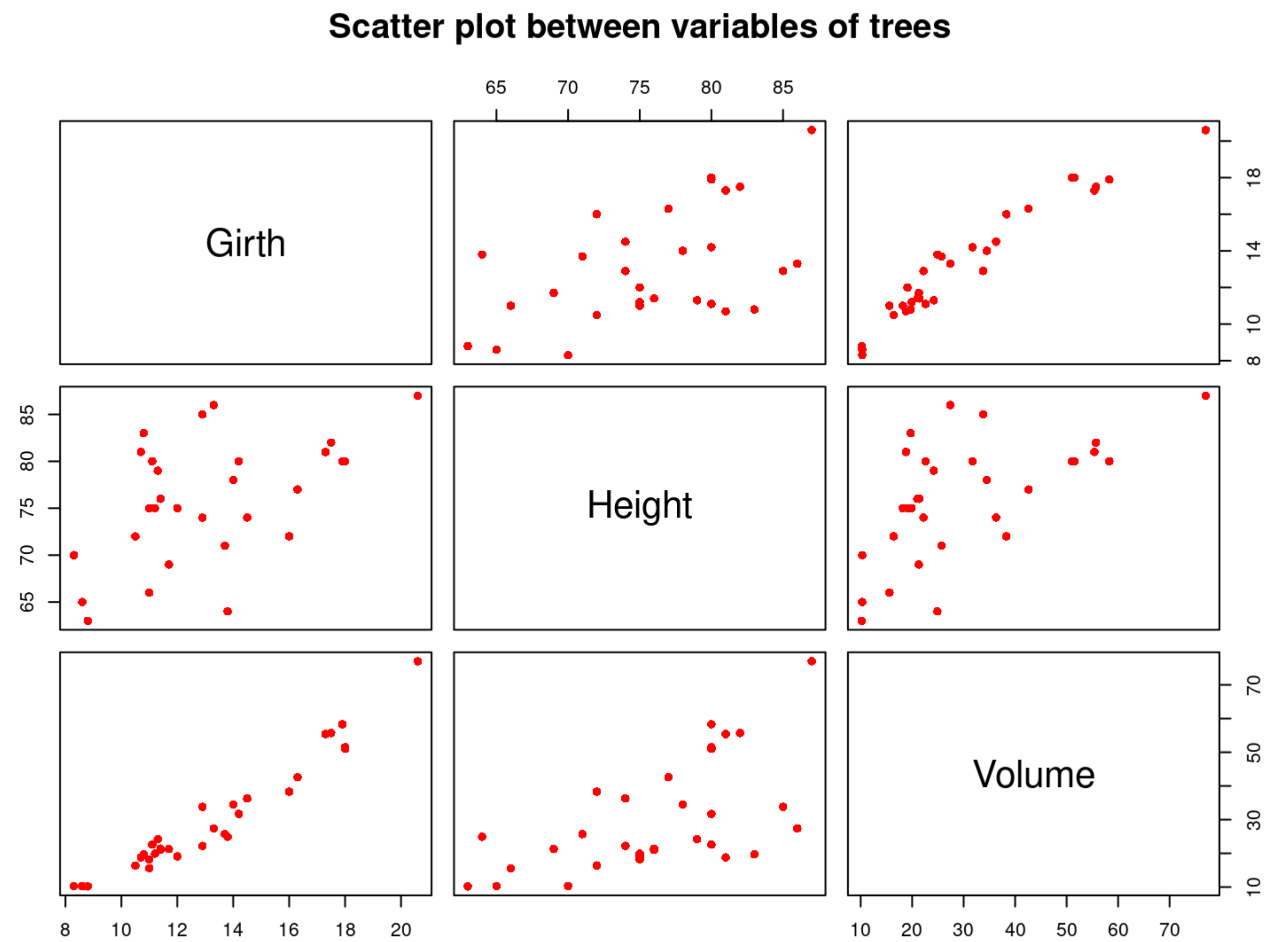
```
summary(trees)
```

```
     Girth           Height       Volume
 Min.   : 8.30   Min.   :63   Min.   :10.20
 1st Qu.:11.05   1st Qu.:72   1st Qu.:19.40
 Median :12.90   Median :76   Median :24.20
 Mean   :13.25   Mean   :76   Mean   :30.17
 3rd Qu.:15.25   3rd Qu.:80   3rd Qu.:37.30
 Max.   :20.60   Max.   :87   Max.   :77.00
```

Visualizing Data with `plot()`

Scatter plot of `trees` variables

```
plot(trees, col = "red", pch = 16,
     main = "Scatter plot between variables of trees")
```



`save.image()` Function

Saves an external representation of R objects to a file.

Syntax

```
save.image(file = "RData", version = NULL, ascii = FALSE, safe = TRUE)
```

Parameters

- `file`: name of file (use `.RData` extension)
- `ascii`: TRUE for ASCII format, `FALSE` for binary (default)

Example

```
save.image(file = "myworkspace.RData")
```

To load later:

```
load("myworkspace.RData")
```

Getting Help in R

Use `?` or `help()`.

```
?summary
help(summary)
?rnorm
help(rnorm)
```

Key Differences

`head()` vs `tail()`

Feature	`head()`	`tail()`
Purpose	Shows records from **start**	Shows records from **end**
Default n	6	6

`edit()` vs `fix()`

Feature	`edit()`	`fix()`
Changes saved?	No (temporary)	Yes (permanent)
Assignment needed?	Yes	No

`class()` vs `typeof()`

Feature	`class()`	`typeof()`
Purpose	Object class/type	Internal data type
Granularity	Higher level	Lower level

Exercise: explore other Datasets